

Trigonometrische Form	Harmonische Form	Komplexe Form
$x(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)$	$x(t) = c_0 + \sum_{n=1}^{\infty} c_n \cos(n\omega_0 t - \varphi_n)$	$x(t) = \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 t}$
$\omega_0 = \frac{2\pi}{T}$	$\omega_0 = \frac{2\pi}{T}$	$\omega_0 = \frac{2\pi}{T}$
$a_0 = \frac{2}{T} \cdot \int_0^T x(t) dt$	$c_0 = \frac{a_0}{2}$	$c_n = \frac{1}{T} \cdot \int_0^T x(t) \cdot e^{-jn\omega_0 t} dt$
$a_n = \frac{2}{T} \cdot \int_0^T x(t) \cdot \cos(n\omega_0 t) dt$	$c_n = \sqrt{a_n^2 + b_n^2}$	$c_n = \frac{a_n - jb_n}{2}$
$b_n = \frac{2}{T} \cdot \int_0^T x(t) \cdot \sin(n\omega_0 t) dt$	$\varphi_n = \arctan \left(\frac{b_n}{a_n} \right)$	$c_{-n} = \frac{a_n + jb_n}{2}$